Random Forest Machine Learning Project

Predicting Medical Insurance Charges

Name: Bandam Swathi | Artificial Intelligence Program, October Batch - B (Unlox)

Project Title

Medical Insurance Cost Prediction using Random Forest

Dataset Name

Medical Cost Personal Dataset (insurance.csv) from Kaggle (kaggle.com/datasets/mirichoi0218/insurance). It has 1338 rows and 7 columns: age, sex, bmi, children, smoker, region and charges.

Problem Statement

The aim is to predict how much medical insurance charge a person will have to pay, using details like age, bmi, smoking habit and region. Charges is a number, so this is a regression problem.

Data Preprocessing Steps

- Loaded the data and checked its shape, data types and basic statistics.
- Checked for missing values. There were none, so nothing was removed or filled.
- Changed sex and smoker into 0 and 1. Changed region into separate 0/1 columns (get_dummies).
- Split the data into X (features) and y (charges).
- Did a train-test split: 80% for training (1070 rows) and 20% for testing (268 rows).

Model Used

Random Forest Regressor from scikit-learn with 100 trees (n_estimators=100) and random_state=42. It builds many decision trees and takes the average of their answers, so the result is more stable.

Evaluation Metrics

Metric	Value	Meaning
MAE	2545.95	Average error of the prediction
MSE	20,935,331.47	Average of squared errors
RMSE	4575.51	Square root of MSE
R2 Score	0.8651	How much of the change the model explains

Key Findings

- Smoker is the most important feature (about 61%). Smokers pay much higher charges.
- BMI is second (about 22%) and age is third (about 13%).
- Children has a small effect (about 2%). Sex and region matter very little (less than 1% each).
- The model explains about 87% of the change in charges on the test data.

Conclusion

Random Forest worked well on this dataset and gave a good R2 score of 0.87. Smoking, bmi and age are the main things that decide the insurance charges. In future, the model can be made better by tuning the settings like number of trees and depth.